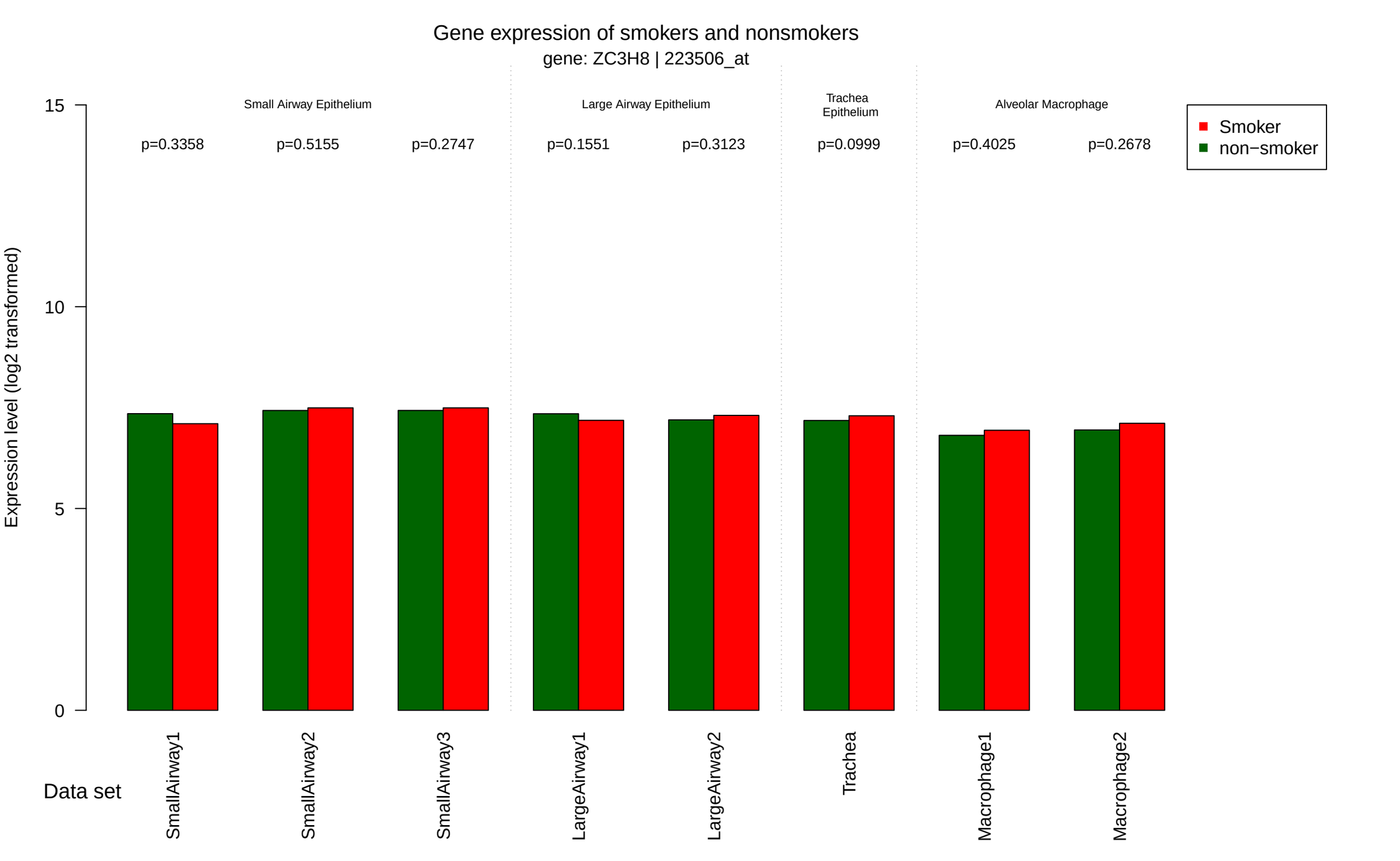
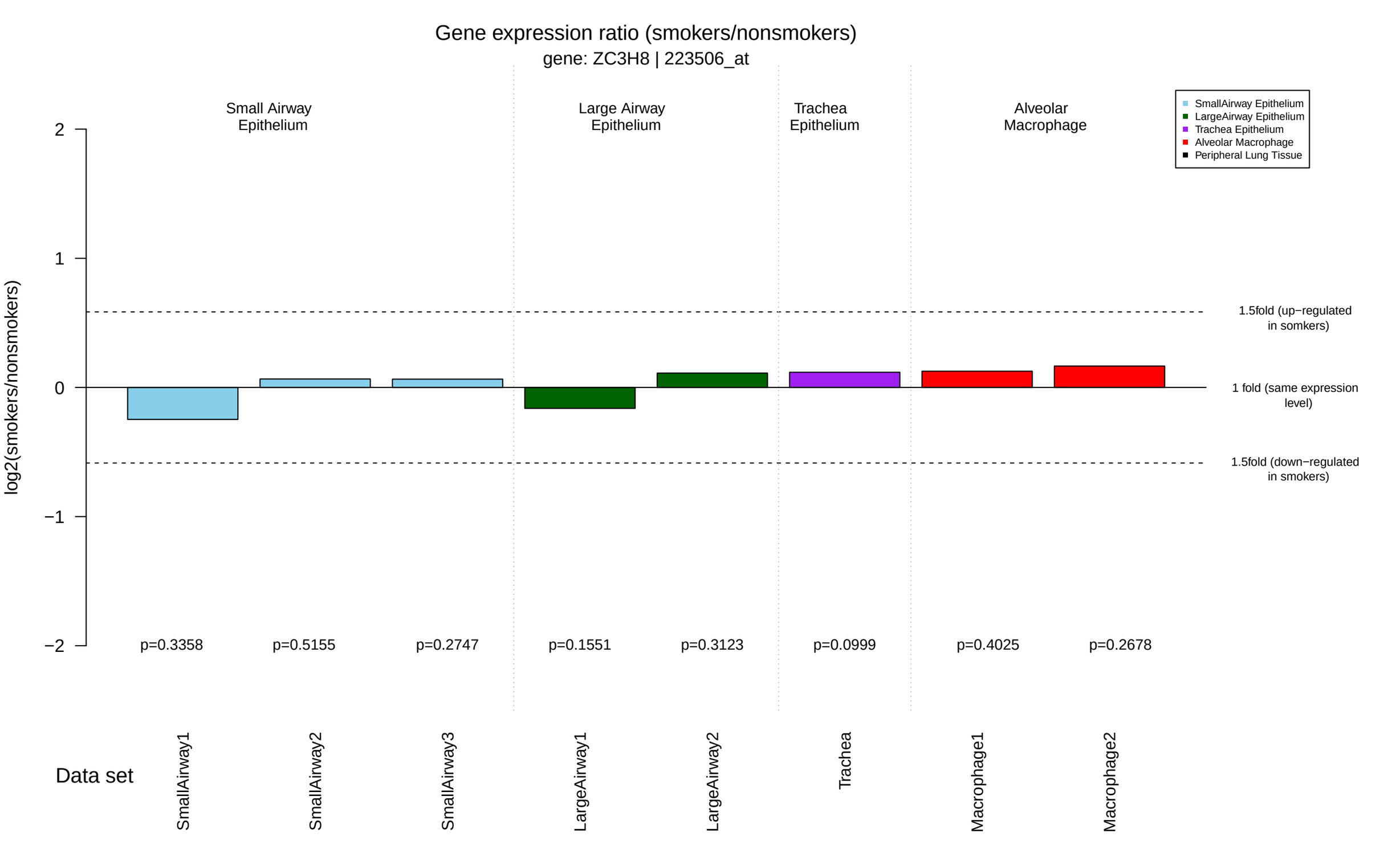
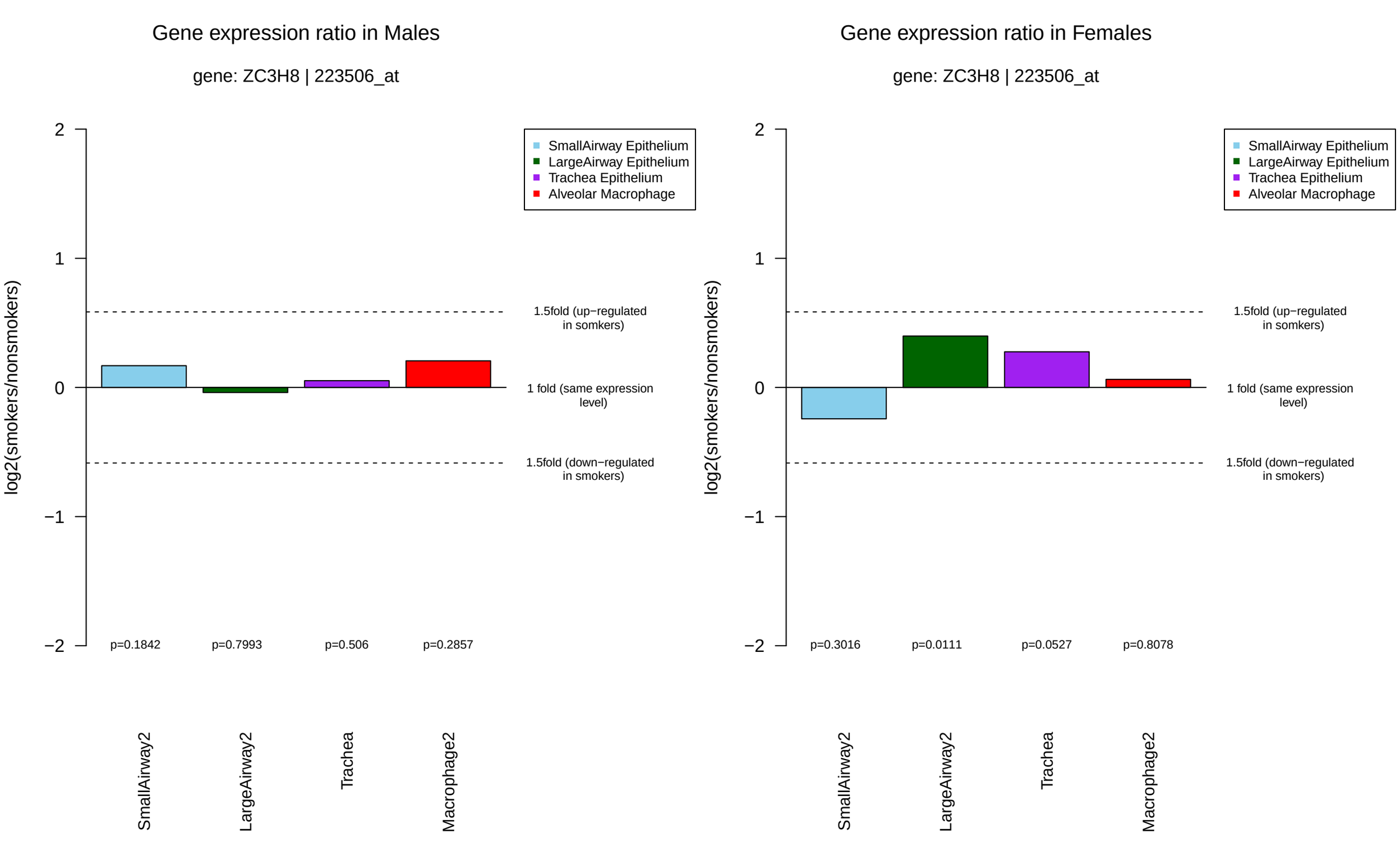
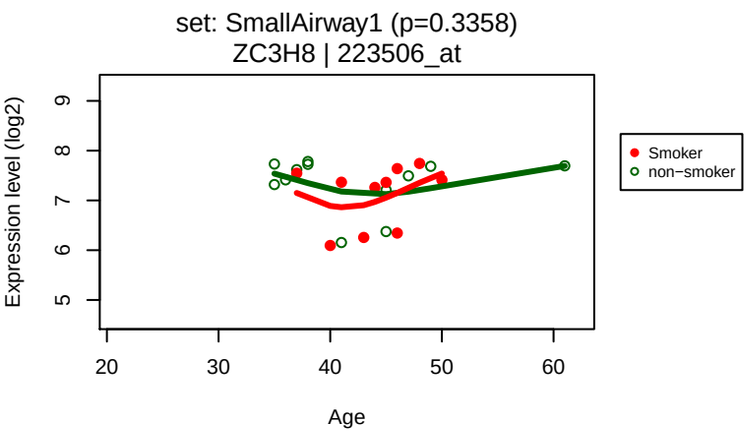
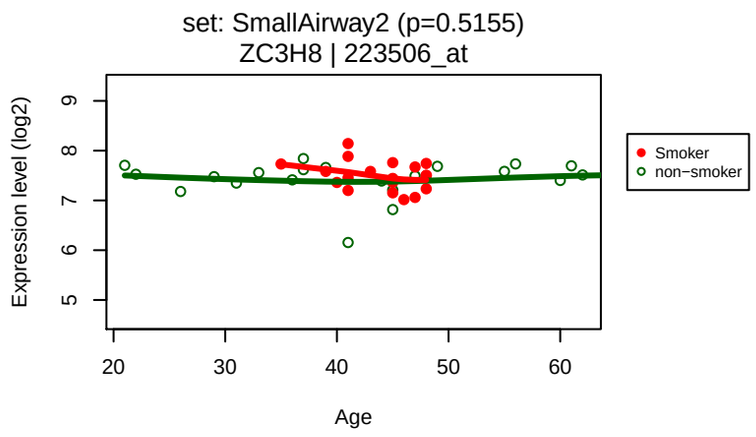
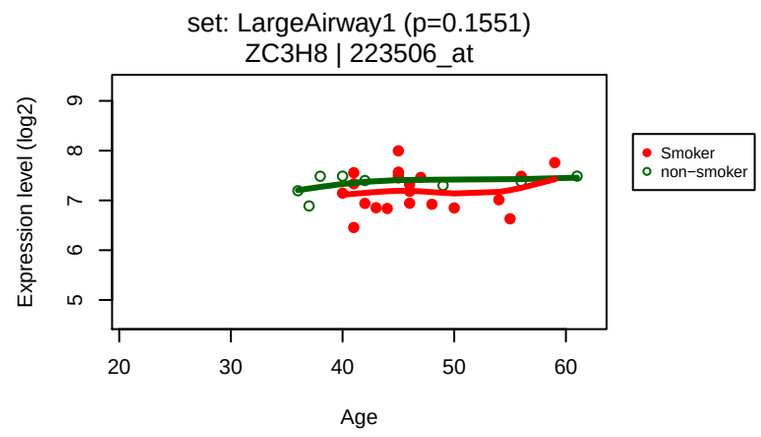

Gene expression of smokers and nonsmokers
gene: ZC3H8 | 223506_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3358
p=0.5155
p=0.2747
p=0.1551
p=0.3123
p=0.0999
p=0.4025
p=0.2678
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

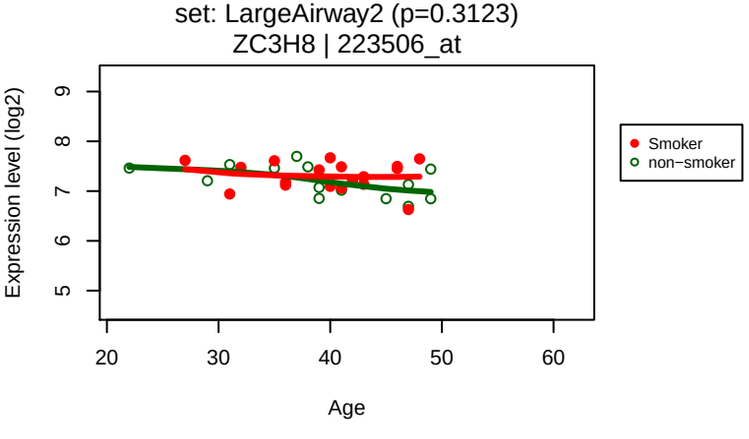

Gene expression ratio (smokers/nonsmokers)
gene: ZC3H8 | 223506_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3358
p=0.5155
p=0.2747
p=0.1551
p=0.3123
p=0.0999
p=0.4025
p=0.2678
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

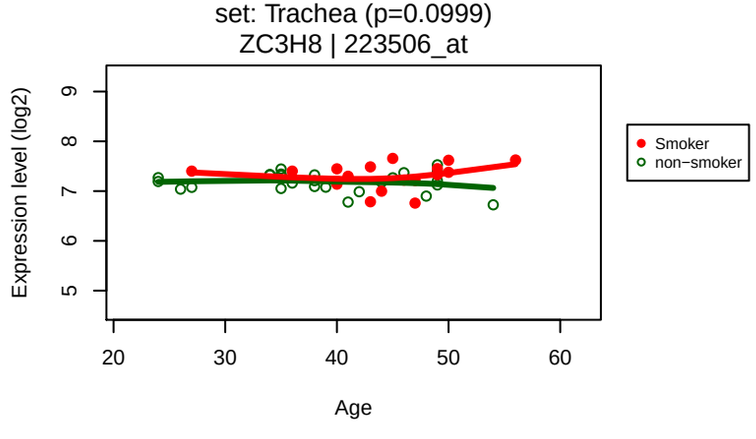

Gene expression ratio in Males
gene: ZC3H8 | 223506_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1842
p=0.7993
p=0.506
p=0.2857
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ZC3H8 | 223506_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3016
p=0.0111
p=0.0527
p=0.8078
SmallAirway2
LargeAirway2
Trachea
Macrophage2

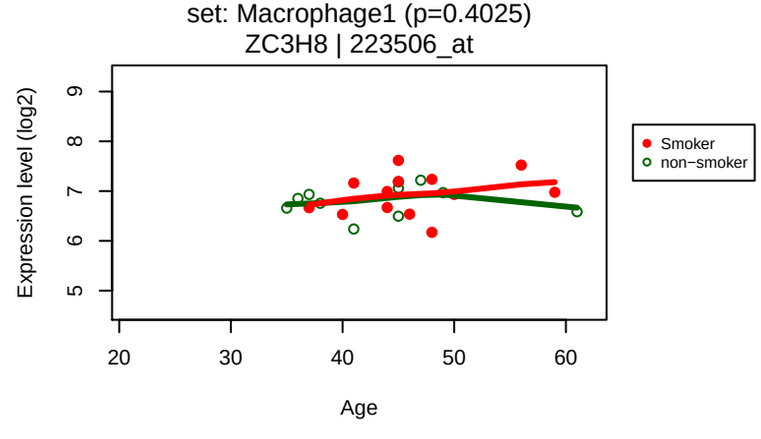

set: SmallAirway1 (p=0.3358)
ZC3H8 | 223506_at
Expression level (log2)
Age
Smoker
non−smoker

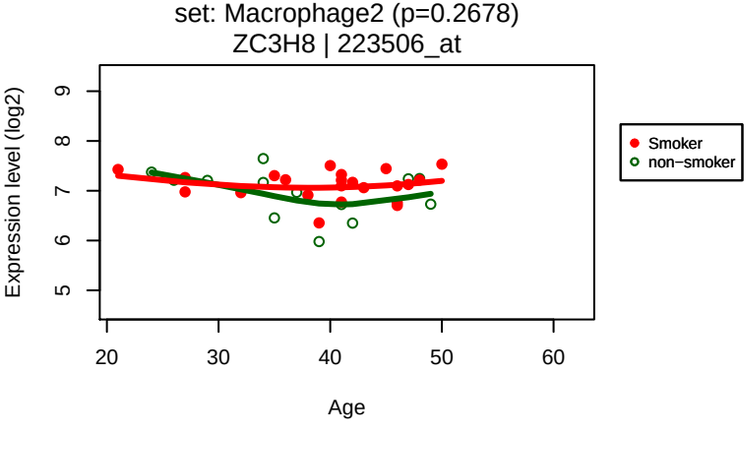

set: SmallAirway2 (p=0.5155)
ZC3H8 | 223506_at
Expression level (log2)
Age
Smoker
non−smoker

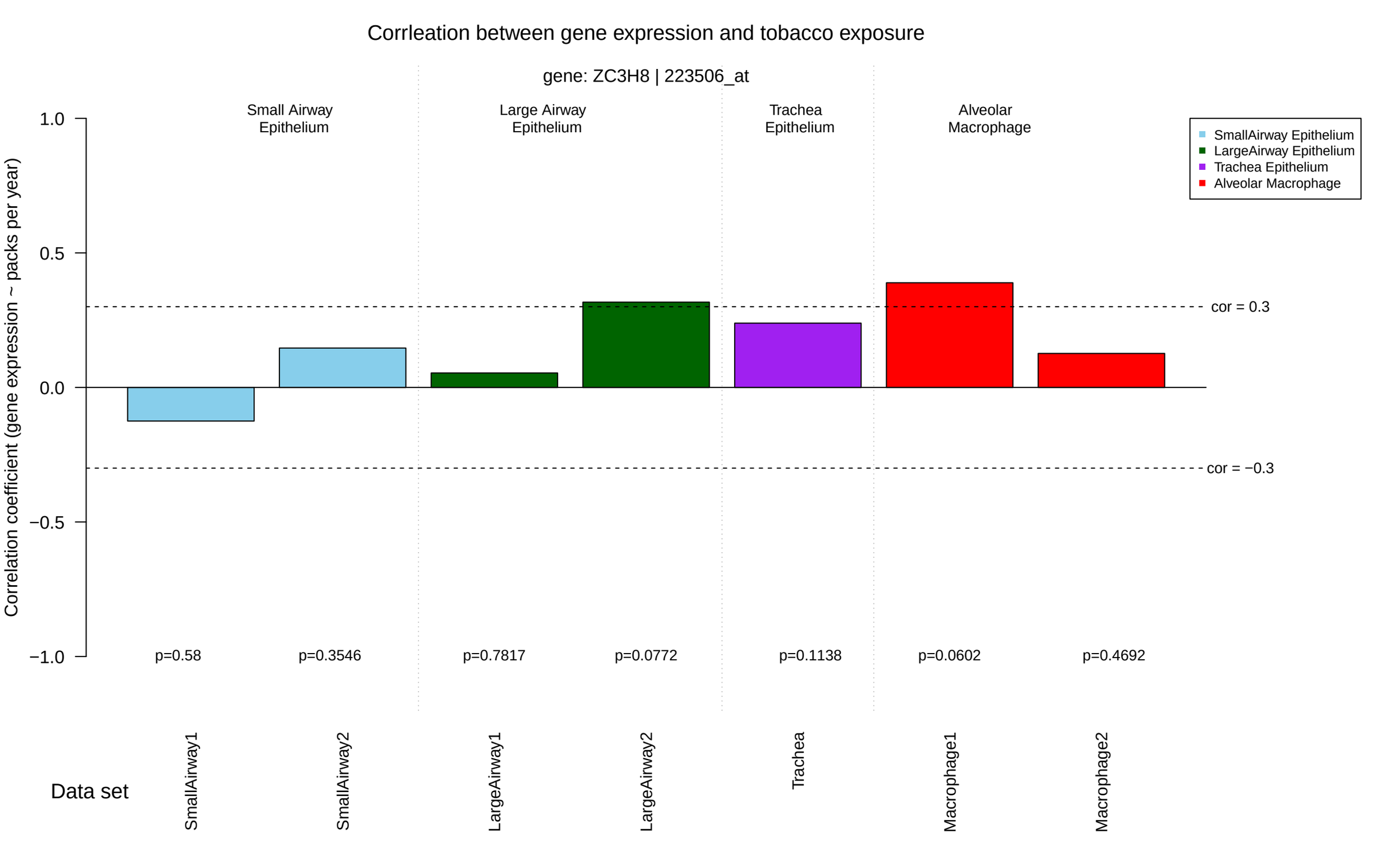

set: LargeAirway1 (p=0.1551)
ZC3H8 | 223506_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3123)
ZC3H8 | 223506_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0999)
ZC3H8 | 223506_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4025)
ZC3H8 | 223506_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2678)
ZC3H8 | 223506_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ZC3H8 | 223506_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.58
p=0.3546
p=0.7817
p=0.0772
p=0.1138
p=0.0602
p=0.4692
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2